

SUMMARY

Fourth-year Data Engineering student focused on turning raw data into reliable pipelines, analytics dashboards, and ML-backed backend services. Through academic and personal projects, I have built end-to-end systems with Python, SQL, BigQuery, Polars, Airflow, FastAPI, MLflow, Docker, and Streamlit, and am seeking a Data Engineer Intern role to apply these skills in a production environment.

EDUCATION

Ho Chi Minh University of Technology Oct. 2022–Jun. 2027 (Expected)
Bachelor of Computer Science, Major in Data Engineering and Big Data Ho Chi Minh City, Vietnam

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript, JavaScript, C++, R
Data Engineering: BigQuery, Polars, Airflow, Star Schema, Supabase/PostgreSQL, Cloudflare R2
MLOps & ML: MLflow (DagsHub), Optuna, Pandera, scikit-learn, XGBoost, Random Forest, LSTM
Backend & AI Apps: FastAPI, Docker, Streamlit, Power BI, LanceDB, SentenceTransformer, RAG, Groq

EXPERIENCE

MyTegami.win May 2025–Jul. 2025
Backend Developer Intern Remote

- Developed the backend AI generation workflow for a bilingual cover letter platform using Genkit and Gemini 2.0 Flash.
- Designed strict Zod schemas to keep outputs structured, reliable, and easier to integrate into the product flow.
- Helped launch support for 14 languages and five writing tones, improving usability for a broader user base.

PROJECTS

Big Data MLOps Movie Recommender System GitHub | Demo
Python, LanceDB, SentenceTransformer, Groq Llama 3.3 70B, RAG, Streamlit, R2, MLflow 2026

- Built the system to solve the cold-explanation problem in recommender systems by combining vector search with an LLM agent that can both recommend movies and explain why.
- Enhanced movie text embeddings by injecting quality tokens and rating signals before encoding MovieLens 25M data into 384D vectors, improving separation between high- and low-quality items.
- Implemented Pseudo-Tower personalization by averaging weighted vectors from user favorites, avoiding a separate graph inference server while preserving personalized retrieval.
- Added a reranker (60% similarity, 30% popularity, 10% quality) and Smart Fallback from Groq function-calling explanations to local LanceDB search for resilient zero-cost deployment.

Scalable Customer Data Platform (CDP) — v2 In Progress GitHub
Python, FastAPI, Optuna, MLflow, Supabase, Cloudflare R2, Pandera, Docker, Airflow, Streamlit Oct. 2025–Present

- Re-architecting a successful Telco churn pipeline into a reusable tabular ML platform after identifying that v1 was too dataset-specific to generalize across business problems.
- Designed v2 around generic CSV ingestion, three-layer column profiling (statistical, semantic, LLM), target suggestion, transform registry, Optuna AutoML, and FastAPI model serving.
- Completed the core infrastructure with Cloudflare R2, Supabase, DagsHub MLflow, ColumnProfile engine, statistical/semantic profiling layers, and reusable transform registry.
- Established v1 baseline results for comparison: **93.5%** accuracy, ROC-AUC **0.99** on 7,043 records, A/B testing service, Streamlit BI dashboard, and estimated ROI of **263x**.

NYC Taxi Data Warehouse & Decision Support System GitHub
Python, BigQuery, Polars, Star Schema, XGBoost, Random Forest, LSTM, Streamlit, Power BI Completed May 2026

- Built the project to practice the full data engineering lifecycle, turning raw NYC TLC taxi trips into a warehouse and ML-driven decision support system for hourly travel demand.
- Modeled a Star Schema with Fact_Trips, Fact_Demand_Hourly feature store, and four dimensions for Time, Location (265 NYC zones), Service Type, and Weather.
- Implemented ETL with Polars and BigQuery to process Parquet trip files and weather CSV data into analytics-ready warehouse tables.
- Trained XGBoost, Random Forest, and LSTM forecasting models, then delivered insights through a Streamlit DSS app and Power BI OLAP dashboard.

LANGUAGES & CERTIFICATES

Languages: English (TOEIC 590) | **Certificate:** HackerRank Software Engineer